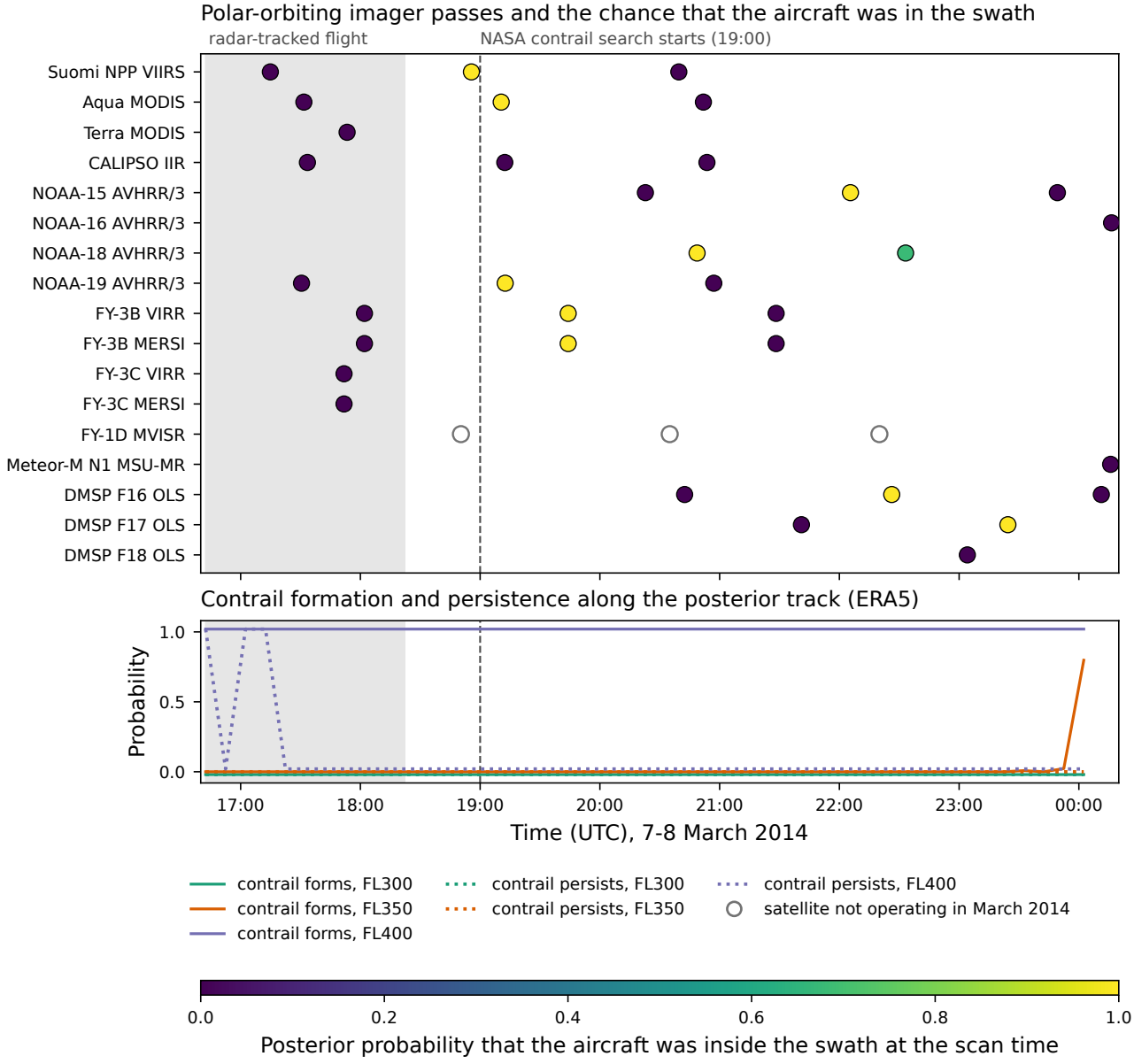

Plain note: each dot is one satellite pass. A yellow dot means the aircraft was almost certainly inside the picture. The lower panel shows that, at 30,000-35,000 ft, the air was too warm for a contrail to form; at 40,000 ft a contrail could form but would not last.

Technical note: posterior = run J, BTO+BFO case, families waypoints 0.9982, free 0.0013, reproduced-radar 0.0003, descent-climb 0.0002 (148,000 routes, 600 s vertices from 18:01:49, linear interpolation); 16:42:27-18:22:12 Geoscience Australia digital military radar track (zero uncertainty). Ephemerides: Space-Track GP elements (Hugging Face export, nearest epoch, SGP4), DMSP F16-F18 from CDAWeb SSJ positions; DMSP F15 not available. Coincidence: zero-Doppler crossing on a sphere (R = 6371 km); swath = maximum scan angle (VIIRS 56.28, MODIS 55.0, AVHRR 55.37, VIRR/MERSI 55.4 deg; OLS 1500 km half-width; IIR 69 km). Swath model checked against granule geolocation to within 5 km. Contrail physics: ERA5 (ARCO mirror) at 300, 238.4, 187.5 hPa; Schmidt-Appleman with overall propulsion efficiency 0.3, emission index 1.25, LHV 43.2 MJ/kg; persistence = ice relative humidity >= 100 %; 1,500-route sample per 10 min. NOAA-18 22:33: 0.68 with the nominal edge, 0.07 with the measured archived (GAC) edge. Lines in the lower panel are offset by 0.02 for legibility.